In [ ]:
```
EXPERIMENT 2 – NUMPY OPERATIONS
PIN NO.: 241UDS0101

AIM:-
To study and practice NumPy operations including array creation, attributes,
indexing, slicing, broadcasting, arithmetic, statistical operations,
concatenation, reshaping, sorting, splitting, and retrieving data.
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [ ]:
```
## Step 2 – Create Arrays for Three Sensors
Values represent readings for Week 1 through Week 4.
```

In [2]:
```
temperature_241UDS0101 = np.array([24, 27, 26, 29])
humidity_241UDS0101 = np.array([62, 58, 65, 61])
soil_moisture_241UDS0101 = np.array([41, 45, 43, 48])

temperature_241UDS0101, humidity_241UDS0101, soil_moisture_241UDS0101
```

Out[2]:
```
(array([24, 27, 26, 29]), array([62, 58, 65, 61]), array([41, 45, 43, 48]))
```

In [ ]:
```
## Step 3 – Check Array Properties
```

In [43]:
```
temperature_241UDS0101.ndim
```

Out[43]:
```
1
```

In [44]:
```
humidity_241UDS0101.shape
```

Out[44]:
```
(4,)
```

In [45]:
```
soil_moisture_241UDS0101.size
```

Out[45]:
```
4
```

In [6]:
```
temperature_241UDS0101.dtype
```

Out[6]:
```
dtype('int64')
```

In [7]:
```
temperature_241UDS0101.itemsize, humidity_241UDS0101.itemsize,
soil_moisture_241UDS0101.itemsize
```

Out[7]:
```
(8, 8, 8)
```

In [ ]:
```
## Step 4 — Retrieve Data Using Indexing and Slicing
```

In [46]:
```
soil_moisture_241UDS0101[0]
```

Out[46]:
```
np.int64(41)
```

In [47]:
```
np.unique(humidity_241UDS0101)
```

Out[47]:
```
array([58, 61, 62, 65])
```

In [48]:
```
temperature_241UDS0101[0:2]
```

Out[48]:
```
array([24, 27])
```

In [49]:
```
humidity_241UDS0101[2:]
```

Out[49]:
```
array([65, 61])
```

In [50]:
```
soil_moisture_241UDS0101[-3:]
```

Out[50]:
```
array([45, 43, 48])
```

In [51]:
```
temperature_241UDS0101
```

Out[51]:
```
array([24, 27, 26, 29])
```

In [ ]:
```
## Step 5 — Assign New Values Using Broadcasting
Copy the temperature array, append two additional readings, then update the
appended values.
```

In [14]:
```
temperature_new_241UDS0101 = temperature_241UDS0101.copy()
temperature_new_241UDS0101
```

Out[14]:
```
array([24, 27, 26, 29])
```

In [16]:
```
temperature_new_241UDS0101 = np.append(temperature_new_241UDS0101, [30, 28])
temperature_new_241UDS0101
```

Out[16]:
```
array([24, 27, 26, 29, 30, 28])
```

In [17]:
```
temperature_new_241UDS0101[4:6] = [31, 29]
temperature_new_241UDS0101
```

Out[17]:
```
array([24, 27, 26, 29, 31, 29])
```

In [ ]:
```
## Step 6 — Arithmetic Operations
```

In [18]:
```
soil_moisture_241UDS0101
```

Out[18]:
```
array([41, 45, 43, 48])
```

In [19]:
```
soil_moisture_241UDS0101 + 5
```

Out[19]:
```
array([46, 50, 48, 53])
```

In [20]:
```
soil_moisture_241UDS0101[0] * 2
```

Out[20]:
```
np.int64(82)
```

In [21]:
```
soil_moisture_241UDS0101[1:3] / 2
```

Out[21]:
```
array([22.5, 21.5])
```

In [ ]:
```
## Step 7 — Statistical Operations
```

In [22]:
```
temperature_241UDS0101, humidity_241UDS0101, soil_moisture_241UDS0101
```

Out[22]:
```
(array([24, 27, 26, 29]), array([62, 58, 65, 61]), array([41, 45, 43, 48]))
```

In [23]:
```
np.sum(temperature_241UDS0101)
```

Out[23]:
```
np.int64(106)
```

In [24]:
```
np.max(temperature_241UDS0101), np.min(temperature_241UDS0101)
```

Out[24]:
```
(np.int64(29), np.int64(24))
```

In [25]:
```
print(np.mean(temperature_241UDS0101), np.mean(humidity_241UDS0101),
np.mean(soil_moisture_241UDS0101))
```

```
26.5 61.5 44.25
```

In [ ]:
```
## Step 8 — Combine All Sensor Data
```

In [26]:
```
temperature_241UDS0101.ndim, humidity_241UDS0101.ndim,
soil_moisture_241UDS0101.ndim
```

Out[26]:
```
(1, 1, 1)
```

In [27]:
```
all_sensor_data_241UDS0101 = np.concatenate([temperature_241UDS0101,
humidity_241UDS0101, soil_moisture_241UDS0101])
all_sensor_data_241UDS0101
```

Out[27]:
```
array([24, 27, 26, 29, 62, 58, 65, 61, 41, 45, 43, 48])
```

In [28]:
```
np.sum(all_sensor_data_241UDS0101)
```

Out[28]:
```
np.int64(529)
```

In [ ]:
```
## Step 9 — Reshape into a 3 × 4 Matrix
```

In [29]:
```
all_sensor_data_241UDS0101.ndim, all_sensor_data_241UDS0101.shape
```

Out[29]:
```
(1, (12,))
```

In [30]:
```
all_sensor_data_241UDS0101 = all_sensor_data_241UDS0101.reshape(3, 4)
all_sensor_data_241UDS0101
```

Out[30]:
```
array([[24, 27, 26, 29],
       [62, 58, 65, 61],
       [41, 45, 43, 48]])
```

In [31]:
```
all_sensor_data_241UDS0101.ndim, all_sensor_data_241UDS0101.shape
```

Out[31]:
```
(2, (3, 4))
```

In [ ]:
```
## Step 10 — Sort the Array
```

In [32]:
```
all_sensor_data_241UDS0101
```

Out[32]:
```
array([[24, 27, 26, 29],
       [62, 58, 65, 61],
       [41, 45, 43, 48]])
```

In [33]:
```
all_sensor_data_241UDS0101[0]
```

Out[33]:
```
array([24, 27, 26, 29])
```

In [34]:
```
np.sort(all_sensor_data_241UDS0101, axis=1)
```

Out[34]:
```
array([[24, 26, 27, 29],
       [58, 61, 62, 65],
       [41, 43, 45, 48]])
```

In [35]:
```
np.sort(all_sensor_data_241UDS0101, axis=0)
```

Out[35]:
```
array([[24, 27, 26, 29],
       [41, 45, 43, 48],
       [62, 58, 65, 61]])
```

In [ ]:
```
## Step 11 — Split the Array
```

In [36]:
```
split_temperature_241UDS0101, split_humidity_241UDS0101,
split_soil_moisture_241UDS0101 = np.split(all_sensor_data_241UDS0101, [1,
2], axis=0)
```

In [37]:
```
split_temperature_241UDS0101
```

Out[37]:
```
array([[24, 27, 26, 29]])
```

In [38]:
```
split_humidity_241UDS0101
```

Out[38]:
```
array([[62, 58, 65, 61]])
```

In [39]:
```
split_soil_moisture_241UDS0101
```

Out[39]:
```
array([[41, 45, 43, 48]])
```

In [ ]:
```
## Step 12 — Retrieve Week 2 Readings of All Sensors
```

In [40]:
```
all_sensor_data_241UDS0101
```

Out[40]:
```
array([[24, 27, 26, 29],
       [62, 58, 65, 61],
       [41, 45, 43, 48]])
```

In [41]:
```
week_2_readings_241UDS0101 = all_sensor_data_241UDS0101[:3, 1:2]
print(week_2_readings_241UDS0101)
```

```
[[27]
 [58]
 [45]]
```

In [ ]:
```
## Step 13 — Load and Display the CSV Dataset
This extra step loads the provided CSV so you can inspect the complete
source dataset.
```

In [42]:
```
greenhouse_data_241UDS0101 =
pd.read_csv('Experiment_2_241UDS0101_Greenhouse_Sensor_Data.csv')
greenhouse_data_241UDS0101
```

Out[42]:

	Week	Temperature_C	Humidity_Percent	Soil_Moisture_Percent	Light_Hours
0	Week1	24	62	41	7
1	Week2	27	58	45	8
2	Week3	26	65	43	6
3	Week4	29	61	48	9

In [ ]:
```
 RESULT:-
NumPy array creation, properties, indexing, slicing, value updates,
arithmetic, statistics, concatenation, reshaping, sorting, splitting, and
selecting Week 2 readings were practiced using a new greenhouse sensor
dataset.
```